AgroBot – Universal AI Based Agricultural Assistant

Presented by:-
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Project Statement

- Farmers often struggle to access timely and accurate agricultural information related to crop selection, pest control, weather conditions, and market prices. This lack of information leads to reduced productivity and poor decision-making.
- AgroBot addresses this problem by providing an AI-powered platform that delivers instant, reliable, and easy-to-understand agricultural guidance. The system uses Artificial Intelligence and Natural Language Processing (NLP) to assist farmers through a simple and accessible interface.

Project Outcomes

- Easy access to agricultural knowledge anytime
- Accurate and context-aware responses to farmer queries
- Reduced information gap between farmers and experts
- Improved decision-making through data-driven insights
- Scalable system supporting multiple crops and regions
- Increased farming efficiency and productivity

Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python, Flask
- **Database:** SQLite, SQLAlchemy
- **APIs:** Gemini API, Weather API
- **AI:** NLP (Chatbot), Computer Vision (Image Analysis)

Modules Implemented

- **User Authentication Module**

  Secure login and registration

  Role-based access (Farmer/Admin)

- **AI Chat Assistant Module**

  Natural language interaction

  Instant agricultural advice

- **Crop Recommendation Module**

  Suggests optimal crops based on conditions

- **Weather Information Module**

  Provides real-time weather updates

- **Market Information Module**

  Displays current crop prices

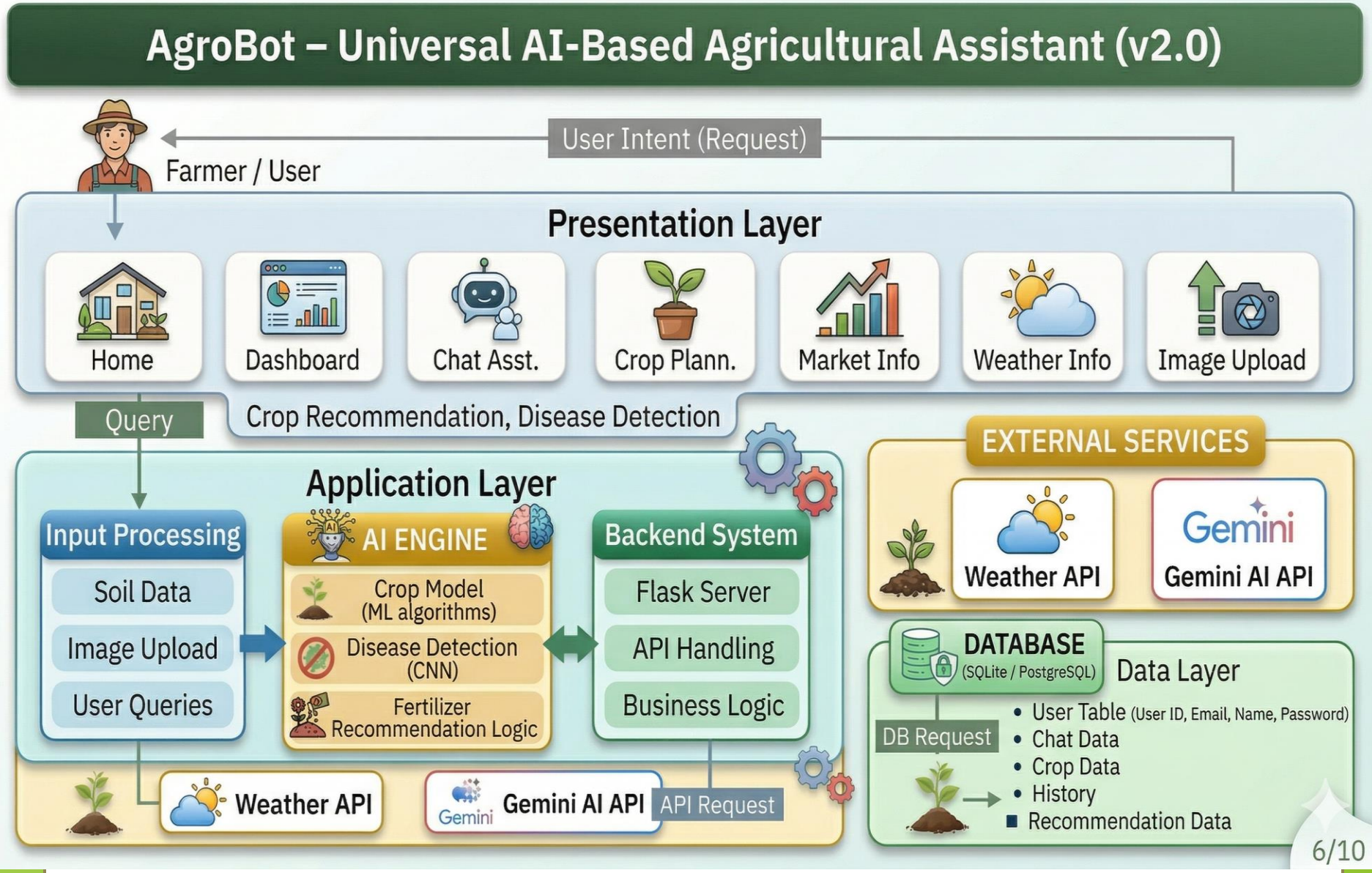
- **Pest & Disease Module**

  Identifies crop diseases and suggests treatments

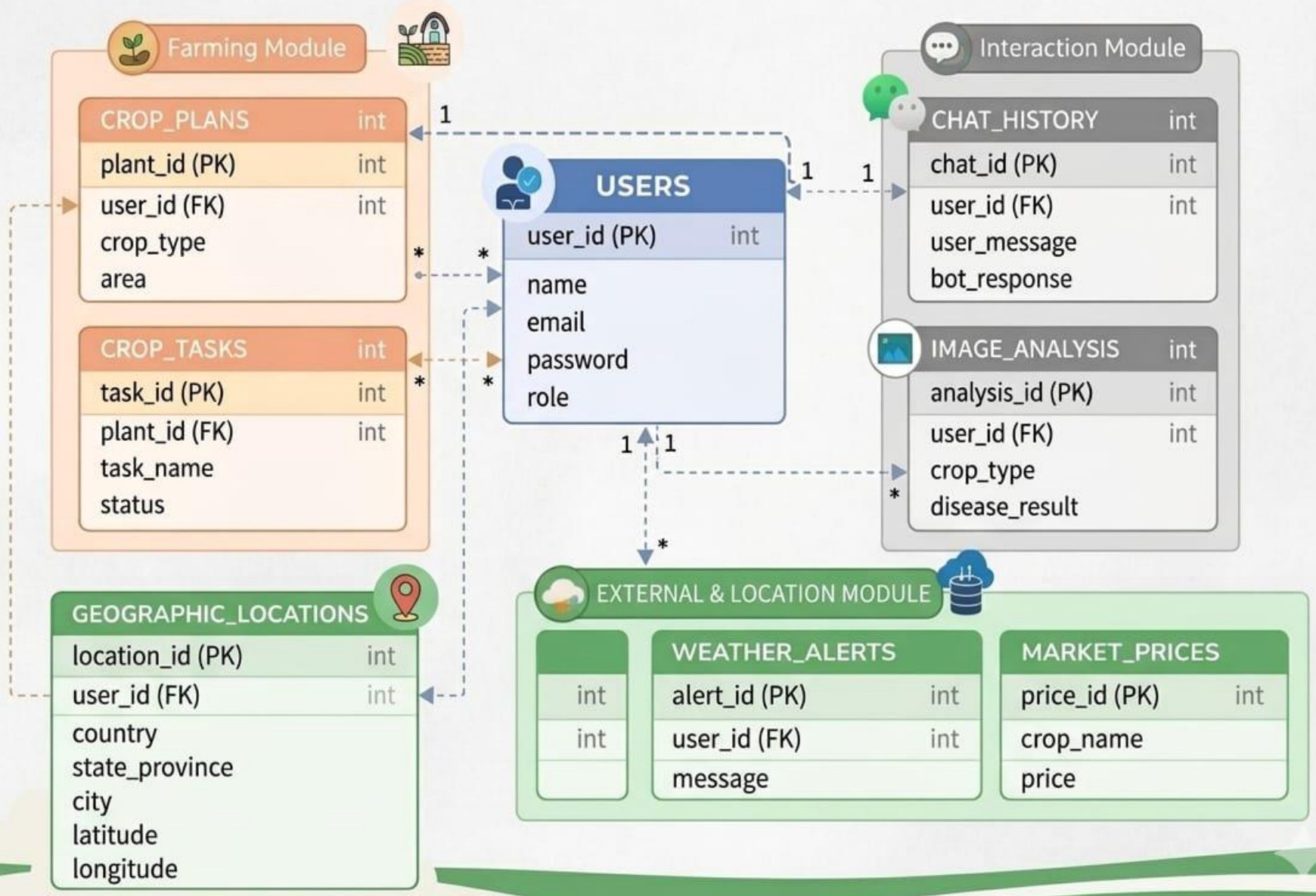
- **Admin Module**

  Manage users and website data

Architecture Diagram



AgroBot — **Database Schema**

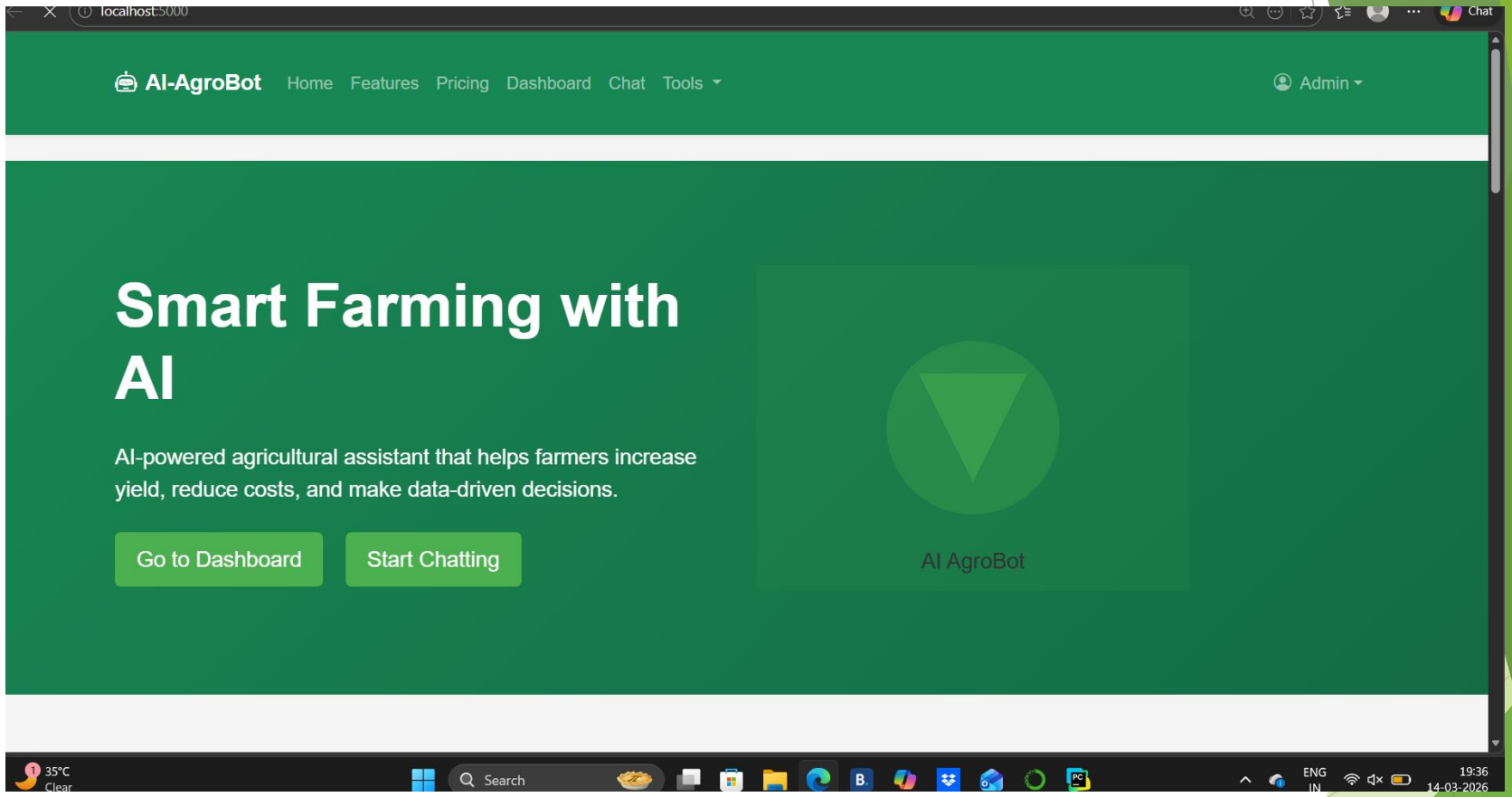


Feature Enhancements

- Integration with government schemes.
- Improve AI API with better version and using different LLMs.
- Mobile application version
- Connecting IOT Device support

Conclusion

- AgroBot is an AI-powered agricultural assistant that simplifies access to farming knowledge. By combining AI, NLP, and real-time data, the system helps farmers make informed decisions regarding crops, diseases, weather, and market conditions.
- This project demonstrates how technology can improve agricultural productivity and support sustainable farming practices.


localhost:5000
Chat
AI-AgroBot
Home
Features
Pricing
Dashboard
Chat
Tools
Admin
Smart Farming with AI
AI-powered agricultural assistant that helps farmers increase yield, reduce costs, and make data-driven decisions.
Go to Dashboard
Start Chatting
AI AgroBot
35°C
Clear
Search
ENG
IN
19:36
14-03-2026

Chat History

ID	User	Question	Actions
19	Arjun	[Image Analy	
18	Arjun	[Image Analy	
17	Arjun	Explain crop	
16	Arjun	hi	
15	Arjun	How to prepa	
14	Arjun	How to prepa	
13	Arjun	[Image Analy	
12	Arjun	tell 2 lines ab	

Chat Details #19

User: Arjun

Time: 2026-03-07 13:58:05

Type: image

Language: en

User Question:

[Image Analysis] Analyze this image

AI Response:

{"health_status": "Likely healthy leaf", "analysis": "Leaf appears healthy with good green coverage.", "green_percentage": 46.0, "recommendations": ["Take multiple photos from different angles", "Check soil moisture levels", "Look for signs of pests on underside of leaves"]}

Uploaded Image:

Chat Details #4

User: Arjun

Time: 2026-03-03 13:54:26

Type: text

Language: en

User Question:

hi

AI Response:

Hello! I'm AgroBot, your agricultural assistant. How can I help you with farming today?